



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

12-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	C	D	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	A,B,D	A,B,C	A,C	A,B	A,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.23 to 2.24	-0.10	15.00	1.50	270.00	-1.25
SECTION-II (ii)	Q.	7	8	9			
	A.	2	4	7			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	A	A	C		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B,C,D	A,B,C	A,B,C,D	A,C	A,B,C,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	9.00	10.00	1.00	4.00	0.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	1	4	8			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	D	B	C	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,D	A,B,D	C,D	A,B,C	A,B,D	A,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.50	0.25	16.00	1305.60	11.80	1.33
SECTION-II (ii)	Q.	7	8	9			
	A.	5	2	11			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

3. Ans (D)

$$x_1 = \frac{1}{2}at_1^2$$

$$x_3 = \frac{1}{2}at_3^2$$

$$(\sqrt{x_3} - \sqrt{x_1}) = (t_3 - t_1) \sqrt{\frac{a}{2}} = (2d) \sqrt{\frac{a}{2}}$$

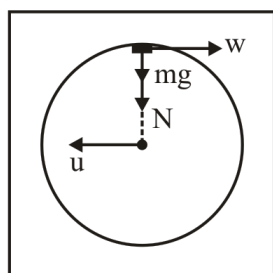
$$(\sqrt{x_3} - \sqrt{x_1}) = d\sqrt{2}\sqrt{a}$$

$$a = \frac{(\sqrt{x_3} - \sqrt{x_1})^2}{2d^2}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (2)



It is elementary to show that the jump of the cube appears at the puck position depicted in the picture on the left hand side. Let u be the cube velocity of mass M at this time moment and let w be the horizontal relative velocity of the puck of mass m with respect to the cube. Since the friction in the system is totally absent, the horizontal projection of the total momentum of the system is conserved.

$$mv = Mu + m(u - w), \dots (i)$$

as well as with the total mechanical energy,

$$\frac{mv^2}{2} = \frac{Mu^2}{2} + \frac{m}{2} [w^2 + (u - w)^2] + 2mgR \dots (ii)$$

In the instant frame of reference associated with the cube, the puck moves with the velocity w along the circle of radius R and its equation of motion projected on the radial direction is given by

$$N + mg = \frac{mw^2}{R} \dots (iii)$$

It is rather obvious that the condition of the cube's jump from the plane of the table is found, according to Newton's third law, as

$$N = Mg \dots (iv)$$

Solving the set of equations (i)-(iv), the puck velocity is obtained as

$$v = \sqrt{gR} \sqrt{5 + \frac{M}{m} + 4\frac{m}{M}} \dots (v)$$

The minimal velocity of the puck is derived from relation (v) by differentiating over M/m ,

$$v_{\min} = 3\sqrt{gR}$$

and it is achieved at the mass ratio

$$M/m = 2.$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

We have, $p(1) - q(1) = 0$, $p(2) - q(2) = 1$, $p(3)$

$$- q(3) = 4$$

$$\Rightarrow p(x) - q(x) - (x - 1)^2 = (x - 1)(x - 2)(x - 3)$$

$$\Rightarrow p(4) - q(4) = 9 + 6 = 15$$

2. Ans (B)

Here,

$$\begin{aligned} S &= \sum_{k=1}^{10} [(10+k)^2 + (9+k)^2 + \dots + (1+k)^2 + k^2] \\ &= \sum_{k=1}^{10} [(10^2 + 9^2 + 8^2 + \dots + 1^2 + 2(10k + 9k + \dots + k) + 11k^2)] \\ &= \frac{10(11)(21)}{6} \times 10 + 110 \sum_{k=1}^{10} k + 11 \sum_{k=1}^{10} k^2 \end{aligned}$$

$$S = 3850 + 110 \times 55 + 11 \times 385$$

$$\frac{S}{55} = 70 + 110 + 77 = 257$$

3. Ans (C)

$$\frac{(a + \frac{1}{a})^2 + (b + \frac{1}{b})^2}{2} \geq \left(\frac{a + \frac{1}{a} + b + \frac{1}{b}}{2} \right)^2 \geq \frac{1}{2} \left(1 + \frac{1}{ab} \right)^2 \left(ab \leq \frac{1}{4} \right)$$

4. Ans (A)

Let the point Q be $(\alpha, 0)$. Also S is $(2, 0)$. Since

$\angle SPQ = 90^\circ$, SQ is a diameter of the circle.

Hence its equation is $(x - \alpha)(x - a) + y^2 = 0$

Since it touches the $y^2 = 8x$,

$(x - \alpha)(x - 2) + 8x = 0$ has equal roots

i.e. $x^2 - x(\alpha - 6) + 2\alpha = 0$ has equal roots

$$\Rightarrow (\alpha - 6)^2 - 8\alpha = 0 \Rightarrow \alpha = 18$$

Hence the equation of the circle is

$$(x - 18)(x - 2) + y^2 = 0 \text{ i.e. } x^2 + y^2 - 20x + 36 = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,D)**

$$\left(\frac{bx + ax - (a^2 + b^2)}{ab} \right) = \left(\frac{bx + ax - (b^2 + a^2)}{(x-a)(x-b)} \right)$$

$$\Rightarrow ((a+b)x - (a^2 + b^2))((x-a)(x-b) - ab) = 0$$

$$\Rightarrow x = \frac{a^2 + b^2}{a+b} = x_2, x = 0 = x_3, x = a+b = x_1$$

$$c = (a+b) - \left(\frac{a^2 + b^2}{a+b} \right) - 0 = \frac{2ab}{a+b}$$

$$\Rightarrow a, c, d \text{ are in H.P.}$$

6. **Ans (A,B,D)**

$$\text{Put } \sin \theta = \frac{2 \tan \theta/2}{1 + \tan^2 \theta/2},$$

$$\cos \theta = \frac{1 - \tan^2 \theta/2}{1 + \tan^2 \theta/2}$$

We get $\tan \theta/2, \tan \phi/2$ are roots of equations

$$(x-a) \left(\frac{1-k^2}{1+k^2} \right) + y \left(\frac{2k}{1+k^2} \right) = a$$

$\Rightarrow \tan \theta/2, \tan \phi/2$ are roots of equations $(x-a)$

$$(1-k^2) + 2ky - a(1+k^2) = 0$$

$\Rightarrow \tan \theta/2, \tan \phi/2$ are roots of equations

$$k^2x - 2yk + 2a - x = 0$$

$$(\text{differential of root})^2 = 4b^2 = \frac{4y^2 + 4x^2 - 8ax}{x^2}$$

$$\Rightarrow y^2 = x^2(b^2 - 1) + 2ax$$

$$\text{roots are } \frac{-2y + \sqrt{4y^2 + 4x^2 - 8ax}}{-2x},$$

$$\frac{-2y - \sqrt{4y^2 + 4x^2 - 8ax}}{-2x} = k$$

$$\Rightarrow \tan \frac{\theta}{2} = \frac{y+bx}{x} \text{ and}$$

$$\tan \frac{\phi}{2} = \frac{y-bx}{x} \left\{ \because \tan \frac{\theta}{2} - \tan \frac{\phi}{2} = 2b \right\}$$

7. **Ans (C,D)**

Any parabola whose axis is parallel to x axis will be of the form

$$(y-a)^2 = 4b(x-c) \quad \dots(i)$$

Now, $\ell x + my = 1$, can be rewritten as

$$y-a = -\frac{\ell}{m}(x-c) + \frac{1-am-\ell c}{m} \quad \dots(ii)$$

(2) will touch (1) if

$$\frac{1-am-\ell c}{m} = \frac{b}{-\frac{\ell}{m}} \Rightarrow -\frac{\ell}{m} = \frac{bm}{1-am-\ell c}$$

$$\Rightarrow c\ell^2 - bm^2 + a\ell m - \ell = 0 \quad \dots(iii)$$

$$\text{but given that } 5\ell^2 + 6m^2 - 4\ell m + 3\ell = 0 \quad \dots(iv)$$

Comparing (3) and (4), we get

$$\frac{c}{5} = \frac{-b}{6} = \frac{a}{-4} = \frac{-1}{3} \Rightarrow c = \frac{-5}{3}, b = 2$$

$$\text{and } a = \frac{4}{3}$$

$$\text{So parabola is } \left(y - \frac{4}{3} \right)^2 = 8 \left(x + \frac{5}{3} \right)$$

Whose focus is $\left(\frac{1}{3}, \frac{4}{3} \right)$ and directrix is $3x + 11 = 0$

8. **Ans (A,B,C)**

Let the middle point of chord be $(t^2, 2t)$. Mid point of chord must lie inside the ellipse

$$\Rightarrow t^4 + 8t^2 - 1 < 0$$

$$\Rightarrow t^2 \in (0, -4 + \sqrt{17}) \quad \dots(1)$$

Also equation of the chord is $T = S_1$

$$\Rightarrow t^2x + 4ty = t^4 + 8t^2$$

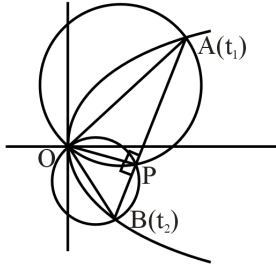
Let this passes through $(\alpha, 0)$

$$\Rightarrow \alpha t^2 = t^4 + 8t^2 \Rightarrow t^4 + (8-\alpha)t^2 = 0 \Rightarrow t^2 = 0$$

$$\text{or } t^2 = \alpha - 8$$

$$\Rightarrow \alpha = t^2 + 8 \Rightarrow \alpha \in (8, 4 + \sqrt{17})$$

9. Ans (A,B,D)



Since OA & OB are diameters of circles

$$\angle OPA = \angle OPB = 90^\circ$$

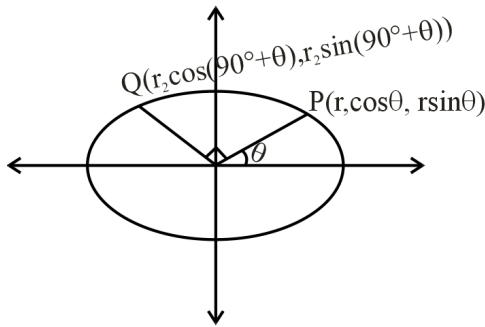
Hence A, P, B are collinear

$$\text{Now } m = \frac{2}{t_1 + t_2} = \frac{2}{\frac{1}{m_1} + \frac{1}{m_2}} = \frac{2m_1 m_2}{m_1 + m_2}$$

$$\left(m_1 = \frac{1}{t_1} \text{ \& } m_2 = \frac{1}{t_2} \right)$$

10. Ans (A,C)

Let $OP = r_1$ & $OQ = r_2$



Now P & Q lie on the ellipse hence

$$r_1^2 \left(\frac{\cos^2 \theta}{16} + \frac{\sin^2 \theta}{9} \right) = 1 \Rightarrow \frac{\cos^2 \theta}{16} + \frac{\sin^2 \theta}{9} = \frac{1}{r_1^2} \quad \dots(1)$$

$$r_2^2 \left(\frac{\sin^2 \theta}{16} + \frac{\cos^2 \theta}{9} \right) = 1 \Rightarrow \frac{\sin^2 \theta}{16} + \frac{\cos^2 \theta}{9} = \frac{1}{r_2^2} \quad \dots(2)$$

$$\text{Now } (1) + (2) \Rightarrow \frac{1}{16} + \frac{1}{9} = \frac{1}{r_1^2} + \frac{1}{r_2^2} = \frac{25}{144}$$

Let equation of chord PQ be $x \cos \alpha + y \sin \alpha = p$, homogenizing the equation of ellipse with this chord gives

$$\frac{x^2}{16} + \frac{y^2}{9} - \left(\frac{x \cos \alpha + y \sin \alpha}{p} \right)^2 = 0$$

As OP & OQ are perpendicular coefficient of $x^2 + \text{coefficient of } y^2 = 0$

$$\Rightarrow \left(\frac{1}{16} - \frac{\cos^2 \alpha}{p^2} \right) + \left(\frac{1}{9} - \frac{\sin^2 \alpha}{p^2} \right) = 0$$

$$\Rightarrow \frac{1}{16} + \frac{1}{9} = \frac{1}{p^2} \Rightarrow p^2 = \frac{144}{25} \Rightarrow p = 12/5$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (0.50)

$y = ax^2 + bx + c$ must be one one

Case-I : We have $c = 1$, and $4a + 2b + c = 0$

$$\Rightarrow 2a + b = \frac{-1}{2}$$

Case-II : We have $c = 0$ and $4a + 2b + c = 1$

$$\Rightarrow 4a + 2b = 1 \Rightarrow 2a + b = \frac{1}{2}$$

2. Ans (0.25)

$$p^2 + p + 3 \in \left[\frac{11}{4}, \infty \right) \Rightarrow -\frac{b}{a} \in \left[\frac{11}{4}, \infty \right)$$

$$\Rightarrow -\frac{b}{2a} \in \left[\frac{11}{8}, \infty \right)$$

$$\text{Case-I } b = -\frac{1}{2} - 2a \Rightarrow 1 + \frac{1}{4a} \in \left[\frac{11}{8}, \infty \right)$$

But $-\frac{b}{2a}$ cannot take value in $(0, 2)$

$$\Rightarrow 1 + \frac{1}{4a} \in [2, \infty) \Rightarrow \frac{1}{4a} \in [1, \infty) \Rightarrow a \in \left(0, \frac{1}{4} \right]$$

$$\text{Case-II } b = \frac{1}{2} - 2a \Rightarrow 1 - \frac{1}{4a} \in [2, \infty) \Rightarrow -\frac{1}{4a} \in [1, \infty) \Rightarrow -4a \in (0, 1] \Rightarrow a \in \left[-\frac{1}{4}, 0 \right)$$

3. Ans (16.00)

$$xy = 16 \text{ at } \left(4t, \frac{4}{t} \right)$$

Equation of normal

$$\left(y - \frac{4}{t} \right) = t^2(x - 4t)$$

$ty - 4 = t^3x - 4t^4$, it passes through P(h, k)

$$\Rightarrow 4t^4 - t^3(h) + kt - 4 = 0 \quad 4 \text{ normal can be drawn}$$

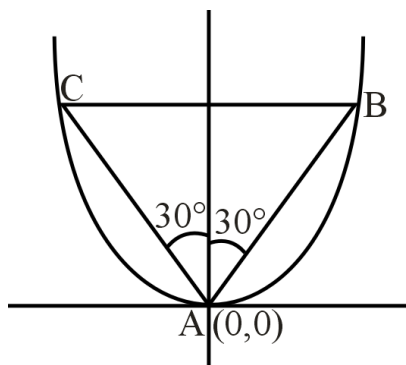
$$(t_1^2 + t_2^2 + t_3^2 + t_4^2) = 4 \left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4} \right)$$

$$\Rightarrow e \left(\sum t_i \right)^2 - 2 \sum t_i t_j = 4 \left(\frac{\sum t_i t_j t_k t_l}{t_1 t_2 t_3 t_4} \right)$$

$$\Rightarrow \frac{h^2}{16} - 2(0) = 4 \frac{\left(\frac{-k}{4} \right)}{\left(\frac{-4}{4} \right)}$$

$$\Rightarrow \frac{h^2}{16} = k \Rightarrow x^2 = 16y$$

4. Ans (1305.60)



Equation of AB is $y = \sqrt{3}x$

Solving it with locus $\Rightarrow x^2 = \sqrt{3}x \Rightarrow x = 16\sqrt{3}, x = 0$

Distance of AB = $32\sqrt{3} \Rightarrow \text{Area} = \frac{\sqrt{3}}{4} \times 1024 \times 3$

$= 768\sqrt{3}$

5. Ans (11.80)

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$= \frac{\sin A \sin B \sin C}{\cos A \cos B \cos C} = \frac{3 + \sqrt{3}}{\sqrt{3} - 1} = 3 + 2\sqrt{3}$$

$$\text{now } \cos(A + B + C) = \cos \pi = -1 = \cos A \cos B \cos C (1 - \Sigma \tan A \tan B)$$

$$\Rightarrow \Sigma \tan A \tan B = 1 + \frac{8}{\sqrt{3} - 1} = 5 + 4\sqrt{3}$$

Now equation whose roots are $\tan A, \tan B, \tan C$ is

$$x^3 - (3 + 2\sqrt{3})x^2 + (5 + 4\sqrt{3})x - (3 + 2\sqrt{3}) = 0$$

$$\Rightarrow \text{roots are } 1, \sqrt{3}, 2 + \sqrt{3}$$

$$\Rightarrow \text{angles} = 45^\circ, 60^\circ, 75^\circ$$

6. Ans (1.33)

$$x^3 - (3 + 2\sqrt{3})x^2 + (5 + 4\sqrt{3})x - (3 + 2\sqrt{3}) = 0$$

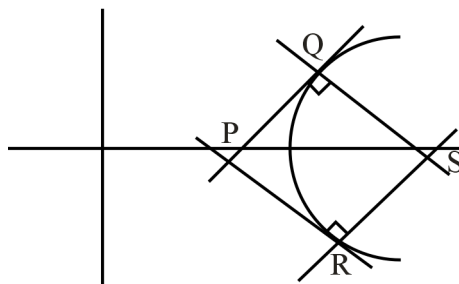
$$\Rightarrow \text{roots are } 1, \sqrt{3}, 2 + \sqrt{3}$$

$$\Rightarrow \text{angles} = 45^\circ, 60^\circ, 75^\circ$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (5)



$\therefore$ S lies on circumcircle of triangle PQR

$\Rightarrow$ parallelogram PQRS is concyclic.

Hence PQRS must be rectangle so $\angle QPR = 90^\circ$

$\Rightarrow$ P must lie on director circle.

$$\Rightarrow \text{locus of P will be } (x - 2)^2 + (y - 1)^2 = 9 - 4$$

$$\Rightarrow x^2 - 4x + 4 + y^2 - 2y + 1 = 5$$

$$\Rightarrow x^2 + y^2 - 4x - 2y = 0 \Rightarrow c = 0, g = -2, f = -1$$

8. Ans (2)

$$S = \sum_{r=1}^{\infty} \frac{r+2}{2^{r+1} \cdot r(r+1)} = \sum \left(\frac{2}{r} - \frac{1}{r+1} \right) \frac{1}{2^{r+1}}$$

$$= \sum \left(\frac{1}{r \cdot 2^r} - \frac{1}{(r+1)2^{r+1}} \right) = \frac{1}{2}$$

9. Ans (11)

$$p(14) = (14 - \alpha)(14 - \beta)(14 - \gamma) = 231$$

$$= 3 \times 7 \times 11$$

$$\Rightarrow (14 - \alpha, 14 - \beta, 14 - \gamma) \text{ can be } (3, 7, 11),$$

$$(-3, -7, 11), (-3, 7, -11), (3, -7, -11),$$

$$(1, -21, -11), (-1, -21, 11), (1, -33, -7),$$

$$(-1, -33, 7), (1 - 3, -77), (-1, 3, -77),$$

$$(1, -1, -231)$$



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ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,D	A,C	A,B,C	A,C	A,D	A,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	C	C	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.25	4	5.90 to 6.10	21.40 to 21.55	-1.28	0.5
SECTION-II (ii)	Q.	7	8	9			
	A.	320	1	24			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	A,B,C	C,D	A,B	B,C,D	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	B	C	C		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	1.00	6.00	1.00	3.00	6.00
SECTION-II (ii)	Q.	7	8	9			
	A.	9	4	4			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,B,C	A,C	A,B,C,D	A,D	A,B,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	D	D	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	24.00	4.00	5.00	0.50	1.00	10.00
SECTION-II (ii)	Q.	7	8	9			
	A.	7	9	0			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

6. Ans (A,D)

Suppose block A and B move together.

Applying NLM on C, A + B and D respectively

$$60 - T = 6a$$

$$T - 18 - T' = 9a \quad T' - 10 = 1a$$

$$\text{Solving } a = 2 \text{ m/s}^2$$

To check slipping between A and B, we have to find friction force in this case. If it is less than limiting static friction, then there will be no slipping between A and B.

Applying NLM on A

$$T - f = 6 \times 2$$

$$\text{As } T = 48\text{N} \quad f = 36\text{N}$$

And $f_{s \text{ max}} = 42\text{N}$ hence A and B move together.

PART-1 : PHYSICS

SECTION-II (i)

3. Ans (5.90 to 6.10)

A plot of the path is shown in figure-(a) with unit vectors $\mathbf{e}_r$ along and $\mathbf{e}_\theta$ perpendicular to $\mathbf{r}$ and in the direction of increasing θ .

(a) **Polar Coordinates:** See figure-(b).

Since $\theta = t^2$, $\dot{\theta} = 2t$; since $r = 2\theta = 2t^2$, $\dot{r} = 4t$

The velocity $\mathbf{v}$ at $\theta = \frac{\pi}{3}$ radians is found as follows:

$$\theta = \frac{\pi}{3} = t^2 \text{ or } t = 1.023 \text{ s}$$

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta = 4(1.023)\mathbf{e}_r + [2(1.047)] [2(1.023)]\mathbf{e}_\theta$$

$$= 4.09\mathbf{e}_r = 4.28\mathbf{e}_\theta$$

$$\text{and } v = \sqrt{(4.09)^2 + (4.28)^2} = 5.92 \text{ m/s with}$$

$$\theta_x = 30^\circ + \tan^{-1} 4.09/4.28 = 73.7^\circ.$$

(b) **Cartesian Coordinates.** See figure-(c)

$$x = r \cos \theta = 2\theta \cos \theta \quad y = r \sin \theta = 2\theta \sin \theta$$

$$= 2t^2 \cos t^2 \quad = 2t^2 \sin t^2$$

Then

$$x = 4t \cos t^2 + 2t^2 (-\sin t^2)(2t) = -1.66 \text{ m/s}$$

$$y = 4t \sin t^2 + 2t^2 (\cos t^2)(2t) = +5.68 \text{ m/s}$$

$$\text{at } t = 1.023 \text{ s } (\cos t^2 = \cos \pi/3, \sin t^2 = \sin \pi/3).$$

Hence

$$v = \sqrt{(-1.66)^2 + (5.68)^2} = 5.92 \text{ m/s with}$$

$$\theta_x = \tan^{-1} \frac{5.68}{1.66} = 73.7^\circ$$

5. Ans (-1.28)

Mass of cavity

$$m = \left(\frac{M}{\frac{4}{3}\pi R^3} \right) \left(\frac{4}{3}\pi \left(\frac{R}{4} \right)^3 \right) = \frac{M}{64}$$

$$\text{Now, } V_T = V_R + V_C$$

Here T stands for total, R for remaining portion and C for cavity.

$$\therefore V_R = V_T - V_C \quad (\text{at point A})$$

$$= -\frac{GM}{R^3} \left[1.5R^2 - 0.5 \left(\frac{R}{2} \right)^2 \right] + \frac{3Gm}{2(R/4)}$$

$$= -\frac{41}{32} \left(\frac{GM}{R} \right)$$

6. Ans (0.5)

From energy conservation,

$$K_A + U_A = K_B + U_B$$

$$\text{or } 0 + mV_A = \frac{1}{2}mv^2 + mV_B$$

$$\text{or } v = \sqrt{2m(V_A - V_B)} \quad \dots(i)$$

At point B :

$$V_R = V_T - V_C$$

$$= -\frac{GM}{R^3} \left[1.5R^2 - 0.5 \left(\frac{R}{4} \right)^2 \right] + \frac{Gm}{(R/4)}$$

$$= -\frac{45}{32} \frac{GM}{R}$$

Substituting in equation (i), we have

$$v = \sqrt{\frac{GM}{4R}}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C)

Let the point of intersection of tangents A and B be P(h, k). Then the equation of AB is

$$\frac{xh}{4} + \frac{yk}{1} = 1 \dots (i)$$

Homogenizing the equation of ellipse using (i), we get

$$\frac{x^2}{4} + \frac{y^2}{1} = \left(\frac{xh}{4} + \frac{yk}{1} \right)^2$$

$$\text{or } x^2 \left(\frac{h^2 - 4}{16} \right) + y^2 (k^2 - 1) + \frac{2hk}{4}xy = 0 \dots (ii)$$

The given equation of OA and OB is

$$x^2 + 4y^2 + axy = 0 \dots (iii)$$

Since (ii) and (iii) represent the same line,

$$\frac{h^2 - 4}{16} = \frac{k^2 - 1}{4} = \frac{hk}{2a}$$

$$\text{or } h^2 - 4 = 4(k^2 - 1)$$

$$\text{or } h^2 - 4k^2 = 0$$

Therefore, the locus is $(x - 2y)(x + 2y) = 0$

2. Ans (A,B,C)

$$T_r = \frac{8r}{4r^4 + 1} = \frac{8r}{(2r^2 + 2r + 1)(2r^2 - 2r + 1)}$$

$$= 2 \left\{ \frac{(2r^2 + 2r + 1) - (2r^2 - 2r + 1)}{(2r^2 + 2r + 1)(2r^2 - 2r + 1)} \right\}$$

$$= 2 \left\{ \frac{1}{2r^2 - 2r + 1} - \frac{1}{2r^2 + 2r + 1} \right\} = 2 \left\{ \frac{1}{v_r} - \frac{1}{v_{r+1}} \right\}$$

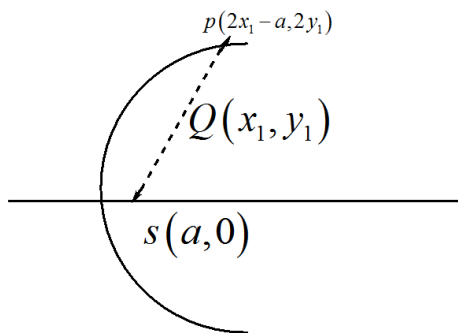
$$\sum_{r=1}^{\infty} T_r = 2 \left[\left(\frac{1}{1} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{13} \right) + \dots \right] = 2$$

$$1 < \sum_{r=1}^n T_r < 2$$

3. Ans (A,C)

Conceptual

4. Ans (A,B,C,D)



Let Q(x₁y₁) be the mid point of SP

⇒ P = (2x₁ - a, 2y₁) lies on

$$y^2 = 4ax$$

$$\Rightarrow 4y_1^2 = a(2x_1 - a) \Rightarrow y_1^2 = 2ax_1 - a^2$$

$$\text{Locus is } y^2 = 2a \left(x - \frac{a}{2} \right)$$

$$\text{Vertex} = \left(\frac{a}{2}, 0 \right)$$

$$\text{Directrix is } x - a/2 = -a/2 \Rightarrow x = 0$$

$$\text{Focus} = (a, 0)$$

5. Ans (A,D)

$$b^2 = (\text{semi minor axis})^2 = \frac{2+3-2}{\sqrt{2}} \cdot \frac{3+5-2}{\sqrt{2}} = 9;$$

$$b = 3$$

6. Ans (A,B,D)

$$\sum \cos(\alpha - \beta) = \frac{-3}{2}$$

$$\Rightarrow \left(\sum \cos \alpha \right)^2 + \left(\sum \sin \alpha \right)^2 = 0$$

$$\Rightarrow \sum \cos \alpha = 0 \quad \& \quad \sum \sin \alpha = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (B)

Sum = 5 ⇒ roots are either 1, 4 or 2, 3

Sum = 6 ⇒ roots are either 1, 5 or 2, 4 or 3, 3

Sum = 8 ⇒ roots are either 1, 7 or 2, 6 or 3, 5 or 4, 4

8. Ans (D)

Sum = 5 ⇒ roots are either 1, 4 or 2, 3

Sum = 6 ⇒ roots are either 1, 5 or 2, 4 or 3, 3

Sum = 8 ⇒ roots are either 1, 7 or 2, 6 or 3, 5 or 4, 4

9. Ans (D)

AB is focal chord

$$\therefore \frac{y_1}{2} \cdot \frac{y_2}{2} = -1 \quad \& \quad x_0 = \frac{x_1 + x_2}{2}, \quad y_0 = \frac{y_1 + y_2}{2}$$

$$A \left(\frac{y_1^2}{4}, y_1 \right)$$

$$B \left(\frac{y_2^2}{4}, y_2 \right) \equiv \left(\frac{4}{y_1^2}, \frac{-4}{y_1} \right)$$

$$C \left(\frac{y_3^2}{4}, y_3 \right)$$

$$D \left(\frac{y_4^2}{4}, y_4 \right)$$

$$\text{slope of BC} = \frac{y_3 - y_2}{x_3 - x_2}$$

$$\text{slope of AE} = \frac{y_1 - y_2}{x_1 - 2}$$

$$\therefore \frac{y_3 - y_2}{x_3 - x_2} = \frac{y_1}{x_1 - 2}$$

$$\frac{4}{y_3 + y_2} = \frac{4y_1}{y_1^2 - 8}$$

$$\Rightarrow 4(y_1^2 - 8) = 4y_1y_3 + 4(-4)$$

$$\Rightarrow y_3 = \frac{4y_1^2 - 16}{4y_1} = y_1 - \frac{4}{y_1} = y_1 + y_2 = 2y_0$$

10. Ans (B)

Parabola $y^2 = 4x$ & $y_1 y_4 = 4$

Tangent at D : $yy_4 = 2(x + x_4)$

Normal at A : $y - y_1 = \frac{-y_1}{2}(x - x_1)$

$\Rightarrow yy_4 - y_1 y_4 = \frac{-y_1 y_4}{2}(x - x_1)$

Abscissa of intersection point is

$$x = 1 + \frac{x_1 - x_4}{2}$$

$$= 1 + \frac{x_1 - \frac{1}{x_1}}{2} \left\{ \begin{array}{l} \text{use : } \frac{1}{x_1} = x_2 \\ 2x_0 = x_1 + x_2 \end{array} \right\}$$

$$= 1 + x_0 - x_2$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (24.00)

P (4, 3) $4x - 3y = 7$

$\frac{x-4}{3} = \frac{y-3}{4} = -\frac{5}{2} \Rightarrow Q\left(\frac{5}{2}, 1\right)$

$\frac{x-4}{3} = \frac{y-3}{4} = -5 \Rightarrow R(1, -1)$

Let $S_1 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

& $S_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$

for $S_1 : \frac{5}{2a^2} - \frac{y}{b^2} = 1$

$\Rightarrow \frac{5}{2a^2} = \frac{4}{7}, \frac{1}{b^2} = 1$

$S_1 : \frac{x^2}{35/8} - \frac{y^2}{7/3} = 1 \Rightarrow e_1^2 : \frac{7/3}{35/8} + 1 \Rightarrow e_1^2 = \frac{23}{15}$

For $S_2 : \frac{x}{A^2} - \frac{y}{B^2} = 1 \Rightarrow A^2 = \frac{7}{4}, B^2 = \frac{7}{3}$

$\frac{x^2}{7/4} + \frac{y^2}{7/3} = 1 \Rightarrow e_2^2 = 1 - \frac{7/4}{7/3} \Rightarrow e_2^2 = \frac{1}{4} \Rightarrow e_2 = \frac{1}{2}$

$S_1 \& S_2 \Rightarrow x^2 \left(\frac{4}{7} + \frac{8}{35} \right) = 2 \Rightarrow y \notin R$

$\Rightarrow 4$ common tangents

2. Ans (4.00)

P (4, 3) $4x - 3y = 7$

$\frac{x-4}{3} = \frac{y-3}{4} = -\frac{5}{2} \Rightarrow Q\left(\frac{5}{2}, 1\right)$

$\frac{x-4}{3} = \frac{y-3}{4} = -5 \Rightarrow R(1, -1)$

Let $S_1 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

& $S_2 : \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$

for $S_1 : \frac{5}{2a^2} - \frac{y}{b^2} = 1$

$\Rightarrow \frac{5}{2a^2} = \frac{4}{7}, \frac{1}{b^2} = 1$

$S_1 : \frac{x^2}{35/8} - \frac{y^2}{7/3} = 1 \Rightarrow e_1^2 : \frac{7/3}{35/8} + 1 \Rightarrow e_1^2 = \frac{23}{15}$

For $S_2 : \frac{x}{A^2} - \frac{y}{B^2} = 1 \Rightarrow A^2 = \frac{7}{4}, B^2 = \frac{7}{3}$

$\frac{x^2}{7/4} + \frac{y^2}{7/3} = 1 \Rightarrow e_2^2 = 1 - \frac{7/4}{7/3} \Rightarrow e_2^2 = \frac{1}{4} \Rightarrow e_2 = \frac{1}{2}$

$S_1 \& S_2 \Rightarrow x^2 \left(\frac{4}{7} + \frac{8}{35} \right) = 2 \Rightarrow y \notin R$

$\Rightarrow 4$ common tangents

3. Ans (5.00)

$a_1 = 1$ and $a_{n+1}^2 = [(1 - \alpha)^2 + \alpha^2] a_n^2$

$= \frac{5}{9} a_n^2 = \left(\frac{5}{9}\right)^n$

Now, $A_n = a_n^2 < \frac{1}{10}$

$\Rightarrow \left(\frac{9}{5}\right)^{n-1} > 10$

$\Rightarrow 9^{n-1} > 2(5^n)$

$\Rightarrow n \geq 5$

4. Ans (0.50)

Diagonal of $(n+1)^{\text{th}}$ square = $\sqrt{2}a_{n+1}$

Now $a_n = \sqrt{2}a_{n+1}$

$\Rightarrow a_n^2 = 2a_{n+1}^2$

$\Rightarrow a_n^2 = 2(2\alpha^2 - 2\alpha + 1)a_n^2$

$\Rightarrow \alpha = 1/2$

5. **Ans (1.00)**

$d > a$; $k = \text{common diff.} \in \mathbb{N}$

$$b + 2k = d = (b - k)^2 + b^2 + (b + k)^2$$

$$= 3b^2 + 2k^2 \geq b + 2k$$

$$\therefore 3b^2 = b \text{ \& } 2k^2 = 2k$$

$$\therefore b = 0 \text{ and } k = 1$$

6. **Ans (10.00)**

$$b = 0, k = 1 \Rightarrow a = -1, c = 1, d = 2$$

$$a + 2b + 3c + 4d = 10$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (7)**

$$\cos(A + B + C) = \cos A \cos B \cos C (1 - \Sigma \tan A \tan B)$$

$$\text{put } A = \alpha, B = 2\alpha, C = 4\alpha$$

8. **Ans (9)**

Let $(2t_1, t_1^2)$ and $(2t_2, t_2^2)$ be the

Extremities of focal chord

$$\Rightarrow t_1 t_2 = -1$$

Equation of focal chord;

$$x(t_1 + t_2) - 2y + 2 = 0 \text{ and the line}$$

$$y = mx + \sqrt{m^2 - \frac{1}{4}} \text{ is}$$

tangent to hyperbola passes through $(0, 1)$

$$\Rightarrow m = \pm \frac{\sqrt{5}}{2} = t_1 + t_2$$

$$\Rightarrow \text{Length of the focal chord} = 9$$

9. **Ans (0)**

Point should be on the segment of $x=2$ line

intercepted between the asymptotes.

$$\alpha \in \left(-\frac{4}{5}, \frac{4}{5} \right)$$